

WTW158 Worksheet on integration 2

Evaluate the following integrals

1. $\int x^4 dx$ 2. $\int (7 - 6x)^4 dx$ 3. $\int \sin^4(5x) \cos 5x dx$

4. $\int \sqrt{x} dx$ 5. $\int x^3 \sqrt{x^4 + 5} dx$ 6. $\int \frac{\sqrt{\ln x}}{x} dx$

7. $\int e^{-5x} dx$ 8. $\int e^{\cos 3x} \sin 3x dx$ 9. $\int e^{3x} \sin(e^{3x}) dx$

10. $\int_{-2}^{-1} \frac{1}{x^4} dx.$

11. $\int 3^x dx$ 12. $\int \frac{4^x}{(1 - 4^x)^6} dx$ 13. $\int \sec^2(2^x) 2^x dx$

14. $\int \operatorname{cosec}(3x) \cot(3x) dx$ 15. $\int x \operatorname{cosec}^2 x^2 dx$ 16. $\int \frac{\sec(\ln x) \tan(\ln x)}{x} dx$

17. $\int 3 dt$ 18. $\int \frac{1}{t^5} dt$ 19. $\int \frac{1}{\sqrt{z}} dz$

20. $\int_1^3 \sin(\pi x) dx.$

21. $\int \frac{2}{1 + 4x} dx$ 22. $\int \frac{2x}{1 + 4x^2} dx$ 23. $\int \frac{2}{1 + 4x^2} dx$

24. $\int \frac{x^4}{1 + x^{10}} dx$ 25. $\int \frac{1}{\sqrt{1 - 9x}} dx$ 26. $\int \frac{1}{\sqrt{1 - 9x^2}} dx$

27. $\int \frac{x^3}{\sqrt{1 - 9x^4}} dx$ 28. $\int \frac{x}{\sqrt{1 - 9x^4}} dx$ 29. $\int \frac{2 + x}{\sqrt{1 - x^2}} dx$

30. $\int_{-2}^{-1} \frac{1}{x} dx.$

31. $\int \frac{t^9 \ln(1 + t^{10})}{1 + t^{10}} dt$ 32. $\int \frac{\cos 2u}{\sin 2u} du$ 33. $\int \frac{\sin(\ln v)}{v} dv$

34. $\int \sec^2(e^{2t}) e^{2t} dt$ 35. $\int 3^{\tan 4z} \sec^2 4z dz$ 36. $\int \left(\frac{3}{x} - \frac{x}{3}\right) dx$

37. $\int \frac{2 + x^4}{x^2} dx$ 38. $\int \frac{x}{2 + x^4} dx$ 39. $\int x^2 \sqrt{x} dx$

40. $\int_{-2}^{-1} |x| dx.$

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